$$A=\begin{bmatrix}1&0&1\\-2&1&0\end{bmatrix}_{2\times 3}\rightarrow \text{SVD}\rightarrow A^T=\begin{bmatrix}1&-2\\0&1\\1&0\end{bmatrix}_{3\times 2}$$

$$A^TA=\underset{3\times 2}{\begin{bmatrix}1&-2\\0&1\\1&0\end{bmatrix}}\underset{2\times 3}{\begin{bmatrix}1&0&1\\-2&1&0\end{bmatrix}}=\begin{bmatrix}5&-2&1\\-2&1&0\\1&0&1\end{bmatrix}_{3\times 3}$$

$$AA^T=\underset{2\times 3}{\begin{bmatrix}1&0&1\\-2&1&0\end{bmatrix}}\underset{3\times 2}{\begin{bmatrix}1&-2\\0&1\\1&0\end{bmatrix}}=\begin{bmatrix}2&-2\\-2&5\end{bmatrix}_{2\times 2}$$

$$\det(A^TA \overset{①}{-} \lambda I)=0 \Rightarrow \begin{bmatrix}5&-2&1\\-2&1&0\\1&0&1\end{bmatrix}-\begin{bmatrix}\lambda&0&0\\0&\lambda&0\\0&0&\lambda\end{bmatrix} \quad ①$$

$$=\begin{bmatrix}5-\lambda&-2&1\\-2&1-\lambda&0\\1&0&1-\lambda\end{bmatrix}\rightarrow(1)\begin{bmatrix}-2&1\\1-\lambda&0\end{bmatrix}+(1-\lambda)\begin{bmatrix}5-\lambda&-2\\-2&1-\lambda\end{bmatrix}=0$$

$$=(1)[0-(1-\lambda)]+(1-\lambda)[(5-\lambda)(1-\lambda)-4]=0$$

$$=-1+\lambda+(1-\lambda)[5-5\lambda-\lambda+\lambda^2-4]=0$$

$$\Rightarrow -1+\lambda+(1-\lambda)[\lambda^2-6\lambda+1]=0$$

$$\Rightarrow \cancel{-1}+\cancel{\lambda}+\lambda^2-6\lambda+\cancel{1}-\lambda^3+6\lambda^2-\cancel{\lambda}=0$$

$$\Rightarrow -\lambda^3+7\lambda^2-6\lambda=0 \Rightarrow \boxed{\lambda^3-7\lambda^2+6\lambda=0}$$

$$\lambda(\lambda^2-7\lambda+6)=0 \longrightarrow \lambda=0$$

$$(\lambda-6)(\lambda-1)=0 \longrightarrow \lambda=6 \quad \lambda=1$$

$$\lambda_1=6,\ \lambda_2=1,\ \lambda_3=0$$

① Si $\lambda=6 \Rightarrow (A^TA-6I)\vec{V}=0$

$$\begin{bmatrix}5-6&-2&1\\-2&1-6&0\\1&0&1-6\end{bmatrix}=\left[\begin{array}{ccc|c}-1&-2&1&0\\-2&-5&0&0\\1&0&-5&0\end{array}\right]\begin{matrix}\rightarrow F_2=F_2+2F_3\rightarrow\\ \rightarrow F_3=F_3+F_1\end{matrix}\left[\begin{array}{ccc|c}-1&-2&1&0\\0&-5&-10&0\\0&-2&-4&0\end{array}\right]$$

$$F_3=F_3-\frac{2}{5}F_2\Rightarrow\left[\begin{array}{ccc|c}-1&-2&1&0\\0&-5&-10&0\\0&0&0&0\end{array}\right]\qquad \begin{aligned}-x-2y+z&=0\\-5y-10z&=0\end{aligned}$$

$$\overset{1}{\cancel{5}}y=-\overset{2}{\cancel{10}}z\Rightarrow y=-2z \qquad \begin{aligned}&x=-2y+z\\&x=-2(-2z)+z\Rightarrow x=4z+z\\&\qquad\qquad\qquad\quad\ \ x=5z\end{aligned}$$

$$(x,y,z)\Rightarrow(5z,-2z,z)\Rightarrow z\underline{(5,-2,1)}$$

② Si $\lambda=1 \Rightarrow (A^TA-I)\vec{V}=0$

$$\left[\begin{array}{ccc|c}4&-2&1&0\\-2&0&0&0\\1&0&0&0\end{array}\right]\begin{matrix}\rightarrow F_2=2F_2+F_1\rightarrow\\ \rightarrow F_3=F_3+\frac{1}{2}F_2\rightarrow\end{matrix}\left[\begin{array}{ccc|c}4&-2&1&0\\0&-2&1&0\\0&0&0&0\end{array}\right]$$

$$\begin{aligned}4x-2y+z&=0\\4x-\cancel{2y}+\cancel{2y}&=0\\4x=0\rightarrow x&=0\end{aligned}\qquad \begin{aligned}-2y+z&=0\\z&=2y\end{aligned}$$

$$(x,y,z)\rightarrow(0,y,2y)$$

$$y(0,1,2)\ ✓$$

③ Si $\lambda=0 \Rightarrow (A^TA-0I)\vec{V}=0$

$$\left[\begin{array}{ccc|c}5&-2&1&0\\-2&1&0&0\\1&0&1&0\end{array}\right]\begin{matrix}\rightarrow F_2=F_2+2F_3\rightarrow\\ \rightarrow F_3=F_3-\frac{1}{5}F_1\rightarrow\end{matrix}\left[\begin{array}{ccc|c}5&-2&1&0\\0&1&2&0\\0&\frac{2}{5}&\frac{4}{5}&0\end{array}\right]$$

$$F_3=F_3-\frac{2}{5}F_2\rightarrow\left[\begin{array}{ccc|c}5&-2&1&0\\0&1&2&0\\0&0&0&0\end{array}\right]\longrightarrow\begin{aligned}5x-2y+z&=0\\y+2z&=0\end{aligned}$$

$$\boxed{y=-2z}\qquad\begin{aligned}&5x-2(-2z)+z=0\\&5x+4z+z=0\\&5x+5z=0\rightarrow \cancel{5}x=-\cancel{5}z=\boxed{x=-z}\end{aligned}$$

$$(x,y,z)\rightarrow(-z,-2z,z)\rightarrow z(-1,-2,1)\ ✓$$

definimos Σ

$$\begin{aligned}\lambda_1=6&\rightarrow\sigma_1=\sqrt{\lambda_1}=\sqrt{6}\\ \lambda_2=1&\rightarrow\sigma_2=\sqrt{\lambda_2}=\sqrt{1}\\ \lambda_3=0&\rightarrow\sigma_3=\sqrt{\lambda_3}=\sqrt{0}\end{aligned}\qquad \Sigma=\begin{bmatrix}\sqrt{6}&0&0\\0&\sqrt{1}&0\end{bmatrix}\ ①$$

Hallar V^* (Vector singular Derecho)

$$\underset{V_1}{\begin{pmatrix}5\\-2\\1\end{pmatrix}}\underset{V_2}{\begin{pmatrix}0\\1\\2\end{pmatrix}}\underset{V_3}{\begin{pmatrix}-1\\-2\\1\end{pmatrix}}\qquad\begin{aligned}|V_1|&=\sqrt{(5)^2+(-2)^2+(1)^2}=\sqrt{30}\\|V_2|&=\sqrt{(0)^2+(1)^2+(2)^2}=\sqrt{5}\\|V_3|&=\sqrt{(-1)^2+(-2)^2+(1)^2}=\sqrt{6}\end{aligned}$$

$$V=\begin{bmatrix}\frac{5}{\sqrt{30}}&0&\frac{-1}{\sqrt{6}}\\-\frac{2}{\sqrt{30}}&\frac{1}{\sqrt{5}}&-\frac{2}{\sqrt{6}}\\1/\sqrt{30}&\frac{2}{\sqrt{5}}&\frac{1}{\sqrt{6}}\end{bmatrix}\rightarrow V^*\rightarrow\begin{bmatrix}\frac{5}{\sqrt{30}}&-\frac{2}{\sqrt{30}}&\frac{1}{\sqrt{30}}\\0&\frac{1}{\sqrt{5}}&\frac{2}{\sqrt{5}}\\-\frac{1}{\sqrt{6}}&-\frac{2}{\sqrt{6}}&\frac{1}{\sqrt{6}}\end{bmatrix}\ ②$$

Hallar U

$$U_1=\frac{1}{\sigma_1}\cdot A\cdot V_1=\frac{1}{\sqrt{6}}\cdot\begin{bmatrix}1&0&1\\-2&1&0\end{bmatrix}_{2\times 3}\begin{bmatrix}\frac{5}{\sqrt{30}}\\-\frac{2}{\sqrt{30}}\\\frac{1}{\sqrt{30}}\end{bmatrix}_{3\times 1}$$

$$=\begin{bmatrix}\frac{6}{\sqrt{30}}\\-\frac{12}{\sqrt{30}}\end{bmatrix}_{2\times 1}\cdot\frac{1}{\sqrt{6}}\Rightarrow\begin{bmatrix}\frac{6}{\sqrt{180}}\\\frac{-12}{\sqrt{180}}\end{bmatrix}=\begin{bmatrix}\frac{\cancel{6}}{\cancel{6}\sqrt{5}}\\\frac{-\overset{2}{\cancel{12}}}{\cancel{6}\sqrt{5}}\end{bmatrix}$$

$$\begin{array}{r|l}\cancel{180}&2\\\cancel{90}&2\\\cancel{45}&3\\\cancel{15}&3\\\cancel{5}&5\\1&\end{array}$$

$$U_1=\begin{bmatrix}\frac{1}{\sqrt{5}}\\-\frac{2}{\sqrt{5}}\end{bmatrix}$$

$$U_2=\frac{1}{\sigma_2}\cdot A\ V_2=\begin{bmatrix}1&0&1\\-2&1&0\end{bmatrix}_{2\times 3}\begin{bmatrix}0\\\frac{1}{\sqrt{5}}\\\frac{2}{\sqrt{5}}\end{bmatrix}_{3\times 1}=\begin{bmatrix}\frac{2}{\sqrt{5}}\\\frac{1}{\sqrt{5}}\end{bmatrix}_{2\times 1}$$

$$U=\begin{bmatrix}\frac{1}{\sqrt{5}}&\frac{2}{\sqrt{5}}\\-\frac{2}{\sqrt{5}}&\frac{1}{\sqrt{5}}\end{bmatrix}\Rightarrow \text{SVD}=U\Sigma V^*$$

$$=\begin{bmatrix}\frac{1}{\sqrt{5}}&\frac{2}{\sqrt{5}}\\-\frac{2}{\sqrt{5}}&\frac{1}{\sqrt{5}}\end{bmatrix}_{2\times 2}\cdot\begin{bmatrix}\sqrt{6}&0&0\\0&1&0\end{bmatrix}_{2\times 3}\begin{bmatrix}5/\sqrt{30}&-\frac{2}{\sqrt{30}}&\frac{1}{\sqrt{30}}\\0&\frac{1}{\sqrt{5}}&-\frac{2}{\sqrt{5}}\\-1/\sqrt{6}&-\frac{2}{\sqrt{6}}&\frac{1}{\sqrt{6}}\end{bmatrix}_{3\times 3}$$